

The empirical recurrence for OEIS A304924, recovered from its tail

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Abstract

OEIS A304924 records “Empirical recurrence of order 96” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 200$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 8$, so the recurrence is proved for every $n \geq 104$. Its last coefficient is $c_{96} = 9 \neq 0$, so no further tail satisfies anything shorter; any order-96 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A304924 is “Number of $n \times 6$ 0..1 arrays with every element unequal to 0, 1, 2, 5 or 7 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

32, 45, 191, 851, 4186, 21214, 104451, 524286, ...

The entry states:

Empirical recurrence of order 96 (see link above)

As of the “Last modified” line on the live entry (May 21 2018 09:15:15, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 200$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 200$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 96: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 8$ is the first whose minimal order is exactly the stated 96.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 400$ exact terms from $n = 8$ on, it returns order 96 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{96} c_i a(n-i),$$

c_1	2	c_{25}	10673239	c_{49}	102759124	c_{73}	−1611540
c_2	11	c_{26}	20217689	c_{50}	31334271	c_{74}	4024163
c_3	52	c_{27}	−17592556	c_{51}	−61749558	c_{75}	3754656
c_4	6	c_{28}	−65196369	c_{52}	−39015713	c_{76}	−76500
c_5	−594	c_{29}	−37104274	c_{53}	16643807	c_{77}	−1845236
c_6	−1706	c_{30}	86862297	c_{54}	26199129	c_{78}	−1090779
c_7	−271	c_{31}	177522215	c_{55}	10562746	c_{79}	176178
c_8	11406	c_{32}	80110399	c_{56}	−19002617	c_{80}	619744
c_9	26015	c_{33}	−168492344	c_{57}	6243309	c_{81}	263882
c_{10}	4700	c_{34}	−302467740	c_{58}	15532015	c_{82}	−69608
c_{11}	−101607	c_{35}	−145486739	c_{59}	17914655	c_{83}	−153895
c_{12}	−246895	c_{36}	213677788	c_{60}	−21880370	c_{84}	−54381
c_{13}	−110320	c_{37}	401725202	c_{61}	−36235610	c_{85}	21365
c_{14}	627387	c_{38}	167367107	c_{62}	−13024316	c_{86}	24468
c_{15}	1472071	c_{39}	−204725085	c_{63}	22922392	c_{87}	9136
c_{16}	816518	c_{40}	−426102595	c_{64}	34867901	c_{88}	−3254
c_{17}	−2329059	c_{41}	−174600848	c_{65}	3567412	c_{89}	−3858
c_{18}	−5877543	c_{42}	192832617	c_{66}	−22559206	c_{90}	−499
c_{19}	−3524407	c_{43}	358375416	c_{67}	−21436047	c_{91}	372
c_{20}	7179992	c_{44}	160745308	c_{68}	5743006	c_{92}	254
c_{21}	13578798	c_{45}	−152182446	c_{69}	17677316	c_{93}	90
c_{22}	4844379	c_{46}	−227610929	c_{70}	7786691	c_{94}	−62
c_{23}	−13008446	c_{47}	−109686970	c_{71}	−6563815	c_{95}	−21
c_{24}	−15156013	c_{48}	100163675	c_{72}	−9498842	c_{96}	9

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 104$, and it is the recurrence OEIS A304924 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 8$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{96} - c_1x^{96-1} - \dots - c_{96}$. Its constant term is $-c_{96} = -9$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 96 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 96, so $\deg q = 96$, and it is monic. It holds from some index t on. From $\max(t, 8)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 96, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 21 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 96, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 9.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-96 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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